

QRPA - compression and PMMs

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Short summary

The idea is to build a QRPA response emulator in two stages. First, we construct a reduced QRPA model by projecting the full QRPA problem onto a response-relevant subspace. Second, we optionally add a small low-rank PMM correction inside this reduced space. In the terminology of Cook's thesis, this is naturally a hybrid emulator: the compressed QRPA model is intrusive and model-driven, while the PMM correction is data-driven and trained only to correct the residual response error.

The guiding picture is

QRPA response $\approx$ dominant response-compressed physical model+small learned low-rank residual.

Problem setting

QRPA response calculations are expensive if they must be repeated many times over a parameter domain. In a typical QRPA setting, one solves a generalized eigenvalue problem

$$NM(\alpha)v_i(\alpha) = E_i(\alpha)v_i(\alpha),$$

where α denotes physical or model parameters, $M(\alpha)$ is the QRPA matrix, N is the QRPA metric, and the eigenpairs give excitation energies and QRPA modes. Given an external field $F(\alpha)$, the transition strengths are

$$B_i(\alpha) = |F(\alpha)^\dagger v_i(\alpha)|^2.$$

The strength function is then obtained by broadening the discrete modes, for instance

$$S(\alpha; \omega) = \frac{\eta}{2\pi} \sum_i \frac{B_i(\alpha)}{(\omega - E_i(\alpha))^2 + (\eta/2)^2}.$$

Many physical observables are integrated response quantities. Examples are

$$m_{-1}(\alpha) = \sum_i \frac{B_i(\alpha)}{E_i(\alpha)}$$

and beta-like phase-space weighted observables of the form

$$O_\beta(\alpha) = \sum_i f_\beta(Q_\beta - E_i(\alpha))B_i(\alpha).$$

Jin et al. [2] recently showed that PMM-type surrogate models can work very well for QRPA linear response problems. Their paper is the natural reference point for the present work. They introduce structured emulators for linear response, including EM1 and EM2, and demonstrate strong results on physical QRPA applications such as electric dipole polarizability and beta-decay half-lives. The comparisons below are only on our simulated QRPA benchmarks. EM1 performs well on the true QRPA data in Jin et al., while we do not yet know whether the response-compressed hybrid will show the same behavior on real QRPA data.

What the EM1 surrogate is

The most relevant comparison point from Jin et al. [2] is their EM1 emulator. EM1 is not a generic feedforward neural network fitted directly to observables. It is a structured PMM-type emulator designed to mimic the spectral form of a QRPA response calculation.

The full QRPA calculation gives excitation energies and transition strengths,

$$\{E_i(\alpha), B_i(\alpha)\}_{i=1}^{N_{\text{modes}}},$$

from which one reconstructs the strength function

$$S(\alpha; \omega) = \frac{\eta}{2\pi} \sum_i \frac{B_i(\alpha)}{(\omega - E_i(\alpha))^2 + (\eta/2)^2}.$$

EM1 replaces the full QRPA problem by a learned low-dimensional parametric matrix problem. At each parameter point α , the emulator constructs an effective matrix

$$\widehat{M}_\theta(\alpha) \in \mathbb{C}^{n_1 \times n_1},$$

diagonalizes it, and interprets the resulting eigenvalues as approximate excitation energies,

$$\widehat{E}_j(\alpha).$$

It also learns an effective transition field or transition-strength model, giving approximate strengths

$$\widehat{B}_j(\alpha).$$

The emulated strength function is then built in the same spectral form as the QRPA response,

$$\widehat{S}_{\text{EM1}}(\alpha; \omega) = \frac{\eta}{2\pi} \sum_{j=1}^{n_1} \frac{\widehat{B}_j(\alpha)}{(\omega - \widehat{E}_j(\alpha))^2 + (\eta/2)^2}.$$

Integrated observables are computed from this reconstructed strength function or from the emulated discrete energies and strengths.

This is why EM1 is a meaningful baseline for our method. It is not just a black-box map

$$\alpha \mapsto S(\alpha; \omega).$$

It keeps the idea that response functions are sums over modes. The main difference is where the reduced model comes from. In EM1, the reduced eigensystem is learned directly as an effective PMM model. In our approach, the reduced eigensystem is obtained by projecting the original QRPA problem onto a response-adapted physical subspace,

$$M_r(\alpha) = B_r^\dagger M(\alpha) B_r, \quad N_r = B_r^\dagger N B_r, \quad F_r(\alpha) = B_r^\dagger F(\alpha),$$

and the PMM part is used as a residual correction,

$$M_{\text{hyb}}(\alpha) = M_r(\alpha) + \Delta_\theta(\alpha).$$

Thus, in Cook’s language, EM1 is a structured PMM emulator for QRPA response, while our method is closer to an intrusive response-compressed reduced basis model augmented by a partially data-driven PMM residual. The difference is not that EM1 needs high-fidelity QRPA data and our method does not. Both methods require exact QRPA calculations at training points. The difference is the type of information used from those calculations. EM1 can be trained from output-level data such as strength functions, observables, and possibly discrete energies and strengths. The response-compressed method needs a richer interface to the QRPA

calculation, especially access to QRPA mode vectors, amplitudes, or matrices, because the basis is built from projectors

$$v_i(\alpha)v_i(\alpha)^\dagger.$$

This is a practical intrusiveness cost rather than a scientific weakness. If the QRPA solver is available, these internal quantities are typically produced, or are close to quantities produced, during the same calculations used to generate the strength functions. The main practical issue is whether they can be exported and stored in a usable form. The advantage is that, when this information is available, the reduced model carries more of the original QRPA structure before any machine learning is introduced.

From spectral compression to response compression

The starting point was the earlier spectral compression idea from the FYS5419/FYS5429 projects. For a parameter-dependent Hermitian problem $H(x)$, one can average projectors onto important eigenspaces. For example, if $P_k(x)$ is the projector onto the lowest k eigenvectors, then

$$Q_{\text{low}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} P_k(x)$$

defines an averaged low-energy projector. The leading eigenvectors of Q_{low} give a reduced basis B_r , and the compressed model is

$$H_r(x) = B_r^\dagger H(x) B_r.$$

For QRPA response functions, the relevant subspace is not necessarily the low-energy eigenspace. What matters is whether a mode contributes to the response. This led us to replace low-energy projector averages with response-weighted projector averages:

$$Q_w = \frac{1}{|\Omega_{\text{train}}|} \sum_{\alpha \in \Omega_{\text{train}}} \sum_i w_i(\alpha) v_i(\alpha) v_i(\alpha)^\dagger.$$

The most basic response weight is

$$w_i^{\text{resp}}(\alpha) = \frac{B_i(\alpha)}{\sum_j B_j(\alpha)}.$$

For polarizability-like quantities, we can instead use

$$w_i^{\text{pol}}(\alpha) = \frac{B_i(\alpha)/E_i(\alpha)}{\sum_j B_j(\alpha)/E_j(\alpha)}.$$

For beta-like observables, we can use

$$w_i^\beta(\alpha) = \frac{f_\beta(Q_\beta - E_i(\alpha)) B_i(\alpha)}{\sum_j f_\beta(Q_\beta - E_j(\alpha)) B_j(\alpha)}.$$

The leading r eigenvectors of Q_w define a response-adapted basis B_r . We then project the QRPA objects as

$$M_r(\alpha) = B_r^\dagger M(\alpha) B_r, \quad N_r = B_r^\dagger N B_r, \quad F_r(\alpha) = B_r^\dagger F(\alpha).$$

This is the intrusive backbone of the method. It uses internal QRPA information, not only input-output data. In Cook's language, this is close to an explicit reduced basis construction, except that the basis is chosen using response relevance rather than only energy or state approximation.

Low-rank PMM correction

The compressed QRPA model can then be corrected by a small PMM-type residual:

$$M_{\text{hyb}}(\alpha) = M_r(\alpha) + \Delta_\theta(\alpha).$$

The philosophy is that $M_r(\alpha)$ should pick up the fundamental physical structure, while the learned term $\Delta_\theta(\alpha)$ should correct what the fixed response-compressed basis misses.

The most interesting correction family so far has been low-rank:

$$\Delta_\theta(\alpha) = U_\theta(\alpha) C_\theta U_\theta(\alpha)^\dagger, \quad U_\theta(\alpha) \in \mathbb{C}^{r \times s}, \quad s \ll r.$$

The parameter dependence is PMM-like but deliberately simple:

$$U_\theta(\alpha) = U_0 + \alpha_1 U_1 + \alpha_2 U_2 + \alpha_1 \alpha_2 U_{12} + \rho_U \sin(\pi \alpha_1) U_s.$$

Here r is the dimension of the compressed QRPA space, while s is the rank of the learned residual correction. This separation is important. The physical reduced model can have dimension $r = 24, 32, 48$, while the learned correction can have rank $s = 1, 2, 4, 8$.

We also tested smaller affine correction models,

$$\Delta_\theta(\alpha) = \sum_{\ell=1}^p f_\ell(\alpha) A_\ell,$$

with small p , simple scalar functions f_ℓ , and symmetric or Hermitian matrices A_ℓ . These are useful as ablations, but the factorized low-rank correction is the cleanest version of the current idea.

To prevent the PMM part from taking over and overfit, we experimented with a penalty

$$\mathcal{L}_\Delta = \frac{1}{|\Omega_{\text{train}}|} \sum_{\alpha \in \Omega_{\text{train}}} \frac{\|\Delta_\theta(\alpha)\|_F^2}{\|M_r(\alpha)\|_F^2}.$$

We also train against response-level quantities rather than matrix error alone. A typical loss is

$$\mathcal{L} = \lambda_S \mathcal{L}_S + \lambda_{m_0} \mathcal{L}_{m_0} + \lambda_{m_{-1}} \mathcal{L}_{m_{-1}} + \lambda_{\text{pol}} \mathcal{L}_{\text{pol}} + \lambda_\beta \mathcal{L}_\beta + \lambda_\Delta \mathcal{L}_\Delta,$$

where

$$\mathcal{L}_S = \frac{1}{|\Omega_{\text{train}}|} \sum_{\alpha} \frac{\|\hat{S}(\alpha; \cdot) - S(\alpha; \cdot)\|_2^2}{\|S(\alpha; \cdot)\|_2^2}.$$

Thus both stages are response-aware: the basis is built from response-weighted projectors, and the correction is trained to reduce response-level residuals.

Cook’s framework

Cook’s thesis gives a useful language for this. In his terminology, emulators can be intrusive, non-intrusive, hybrid, or adaptive depending on how much information they use from the high-fidelity model [1]. PMMs are flexible in this respect. They can be fully data-driven, partially data-driven, or strongly model-driven depending on the problem and available information.

Our method sits naturally in the hybrid region. The compressed QRPA backbone is intrusive because it uses $M(\alpha)$, N , $F(\alpha)$, QRPA modes, and transition strengths. The PMM correction is data-driven because its parameters are trained from response errors. The whole method is therefore a partially data-driven PMM emulator built on top of an intrusive response-compressed reduced basis model.

This is also the conceptual difference from a black-box emulator. A purely data-driven PMM or neural model has to learn the response structure from scratch. Here, the QRPA model first tells us which subspace is physically relevant. The PMM only learns the remaining residual.

Synthetic QRPA experiments

We tested the method in two stages. The first stage used Jin-like synthetic QRPA data. These experiments were designed to have the same mathematical structure as QRPA linear response, but they were still relatively idealized. In this setting, the full QRPA dimension was

$$N_{\text{full}} = 160.$$

We compared compressed QRPA models with reduced dimensions

$$r \in \{8, 12, 16, 24, 32, 48\}$$

against EM1-style PMM surrogates with latent dimensions

$$n_1 \in \{8, 12, 16, 24, 32, 48\}.$$

The dimension-matched comparison is not perfect, since r is a projected physical QRPA subspace while n_1 is a learned latent PMM dimension, but it is still a useful way to compare model sizes. In the early Jin-like benchmark, response compression was much more accurate than the EM1-style surrogate. At $r = n_1 = 32$, a locked comparison gave approximately:

$$Q_{\text{resp}} : \quad \bar{\epsilon}_{S,2} = 0.0130, \quad \text{online speedup} \approx 44.8, \quad N_{\text{break}} \approx 111.$$

The beta-weighted basis gave

$$Q_{\beta} : \quad \bar{\epsilon}_{S,2} = 0.0143, \quad \text{online speedup} \approx 52.2, \quad N_{\text{break}} \approx 109.$$

The best accuracy-optimized low-rank hybrid gave

$$Q_{\beta} + \Delta : \quad \bar{\epsilon}_{S,2} = 0.00762, \quad \text{online speedup} \approx 53.6, \quad N_{\text{break}} \approx 2193.$$

A more interpretable residual version gave

$$Q_{\text{resp}} + \Delta : \quad \bar{\epsilon}_{S,2} = 0.00941, \quad \frac{\|\Delta\|_F}{\|M_r\|_F} \approx 0.029, \quad \text{online speedup} \approx 60.5, \quad N_{\text{break}} \approx 2196.$$

The EM1-style surrogate at $n_1 = 32$ gave

$$\text{EM1} : \quad \bar{\epsilon}_{S,2} = 0.132, \quad \text{online speedup} \approx 129, \quad N_{\text{break}} \approx 3883.$$

This is a useful comparison, but it must be interpreted carefully. EM1 had the highest online speedup, as expected, because evaluating and diagonalizing the learned $n_1 \times n_1$ model is very cheap. Its weakness in this synthetic benchmark was accuracy and break-even cost. In Jin et al.'s actual QRPA applications, however, EM1 performs well. We do not yet know whether our method will show the same advantage on their true data.

The second stage used a more physically constrained synthetic QRPA benchmark. This generator included two-quasiparticle-like energies, structured residual interactions, collective low-rank components, structured transition fields, avoided crossings, QRPA stability checks, and polarizability-like and beta-like observables. The full dimension was again

$$N_{\text{full}} = 160.$$

This benchmark was deliberately harder and more physical.

The results were more modest, but also more informative. Pure response compression remained robust and clearly outperformed the EM1-style output surrogate. At $r = 32$, pure Q_{resp} gave a mean strength-function error of roughly

$$0.145,$$

Method	Dimension	Strength error	Online speedup	Break-even scans
Q_{resp}	$r = 32$	0.0130	44.8	111
Q_{β}	$r = 32$	0.0143	52.2	109
$Q_{\beta} + \Delta$ accuracy	$r = 32, s = 4$	0.00762	53.6	2193
$Q_{\text{resp}} + \Delta$ residual	$r = 32, s = 4$	0.00941	60.5	2196
EM1-style surrogate	$n_1 = 32$	0.132	129	3883

Table 1: Locked comparison in the first Jin-like synthetic QRPA benchmark. The comparison is dimension-matched at $r = n_1 = 32$, although these dimensions have different meanings. In the compression models, r is the dimension of a projected QRPA subspace. In EM1, n_1 is the dimension of a learned effective PMM eigensystem. EM1 has the largest online speedup, while the response-compressed and hybrid models have much lower strength-function error in this simulated setting. This should not be read as a claim about the original Jin et al. physical datasets, where EM1 performs well and where our method has not yet been tested.

while a controlled low-rank hybrid improved this to about

$$0.135.$$

At $r = 48$, pure Q_{resp} gave roughly

$$0.049,$$

while the hybrid improved this to about

$$0.039.$$

So the hybrid improvement survived in a harder regime, but it was less dramatic than in the first toy model.

The dimensional behavior is important. In the physical synthetic benchmark, increasing the compressed dimension from $r = 32$ to $r = 48$ helped significantly. This suggests that, in the harder setting, some of the error at $r = 32$ is still missing subspace dimension, not only a small residual inside the reduced space. The hybrid is therefore most naturally viewed as a refinement of a good response-compressed basis, not as a substitute for choosing a sufficiently rich basis.

Controlled ablations

The most important controlled experiment was the backbone ablation. We compared

$$M_{\text{hyb}}(\alpha) = M_r(\alpha) + \Delta_{\theta}(\alpha)$$

against a low-rank correction trained without the response-compressed QRPA backbone. The result was very clear. The compressed model alone had strength-function error around

$$0.145.$$

The hybrid improved it to about

$$0.135.$$

But the no-backbone correction had error of order

$$1.$$

This shows that the PMM correction is not secretly doing the whole job. The response-compressed QRPA model supplies the structure, and the learned PMM term refines it.

The rank ablation also supported the residual interpretation. With the response-compressed basis fixed, useful improvements appeared already at small correction ranks such as

$$s = 1 \quad \text{or} \quad s = 2.$$

Increasing the correction rank beyond this did not monotonically improve performance and could hurt generalization. This is exactly the pattern one would hope for if the missing component is genuinely a small residual subspace.

The correction-size diagnostics are therefore central. In every experiment we track quantities like

$$\frac{\|\Delta_\theta(\alpha)\|_F}{\|M_r(\alpha)\|_F}, \quad \frac{\|\Delta_\theta(\alpha)\|_F}{\|M_{\text{hyb}}(\alpha)\|_F},$$

the effective rank of Δ_θ , and QRPA stability diagnostics such as imaginary eigenvalue components or negative excitation energies. A good hybrid model should improve the response while keeping these quantities small.

Current interpretation

The current interpretation is:

$$\text{compression gives the physics,} \quad \text{PMM learns the residual.}$$

In Cook’s terminology, the method is not a purely intrusive reduced basis method and not a purely data-driven PMM. It is a hybrid emulator. The response-compressed QRPA model uses internal QRPA information to construct a physically meaningful reduced problem, while the PMM correction learns the remaining response-level error inside that reduced space.

The extra intrusiveness should be viewed as a practical data-access requirement rather than a conceptual weakness. The method does not necessarily require more high-fidelity QRPA solves than an output-level emulator such as EM1. Both approaches need exact QRPA calculations at training points. The difference is that EM1 can often be trained from saved output-level data, such as strength functions, observables, and possibly discrete energies and strengths, while the response-compressed method needs richer information from the same calculations, especially QRPA amplitudes, mode vectors, or matrices. In practice, this means that the method requires a QRPA solver interface that can export the internal objects needed to build response-weighted projectors.

This distinction is important. If only output data are available, then EM1 or EM2-style surrogates are the natural methods. If QRPA internals are available, then the response-compressed method can exploit more of the physics before any learning is introduced. The added intrusiveness is exactly what allows the method to build a response-adapted reduced subspace instead of learning the reduced spectral model entirely from outputs.

The main synthetic evidence so far is:

- Response-weighted compression is much more appropriate for QRPA response functions than ordinary low-energy compression.
- In simulated QRPA settings, response compression is more accurate and more data-efficient than the EM1-style surrogate we implemented.
- EM1 has higher online speedup and can work with less intrusive output-level data, but in these synthetic tests its accuracy is substantially worse and its break-even cost is higher.
- Low-rank PMM corrections can improve the response-compressed model.
- The best corrections are small and low-rank.

- The correction fails without the compressed QRPA backbone, which supports the interpretation that the PMM is not replacing the physics model.

The main open question is whether this carries over to real QRPA data. The ideal dataset would include

$$M(\alpha), \quad N, \quad F(\alpha), \quad v_i(\alpha), \quad E_i(\alpha), \quad B_i(\alpha).$$

If only strength functions and observables are available, then EM1/EM2-style output emulators can still be tested, but the full response-weighted projector method cannot be tested properly. If QRPA matrices or modes are available, then the decisive question is whether real QRPA response manifolds also admit the structure

dominant response-adapted reduced subspace + small low-rank learnable residual.

A fair way to summarize the tradeoff is:

EM1: less intrusive, faster online, output-oriented,

while

response-compressed hybrid: more intrusive, more structured, (so far) substantially more accurate.

Thus the methods are not direct replacements in every data regime. They occupy different points in the emulator taxonomy. EM1 is the natural choice when only output-level data are available. The response-compressed hybrid is the natural choice when one has access to the QRPA solver or can export the internal quantities produced by the solver.

If the real-data tests support the synthetic results, the main claim would be something like:

QRPA response functions can be emulated by first constructing an intrusive response-adapted reduced basis from transition-strength-weighted projectors, and then learning a small low-rank PMM residual correction inside this reduced QRPA space. The resulting method is a hybrid emulator in which the compressed QRPA model carries the dominant physical structure and the PMM correction accounts for residual response-level error.

This is the cleanest version of the idea. It avoids presenting the PMM as a black-box replacement and instead uses it as a controlled correction to an already physical reduced QRPA model.

References

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